

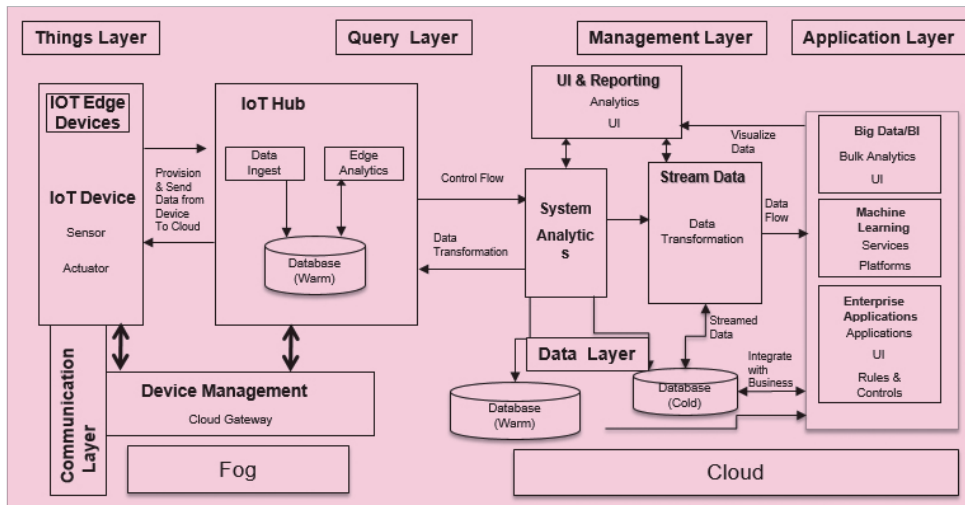


Figure 1: Layers of IoT architecture with databases

- Synchronising data between edge servers and the cloud
 - Real-time data streaming and analytics
 - Data filtering and aggregation
 - Increasing cost of ownership for the database landscape
 - Increased complexity integration and managing the databases to achieve IoT solutions
 - Multiple databases with duplicate functionality and products that are under-utilised
 - Need for a wide range of skills to support the IoT landscape
- Selection of open source databases for IoT implementation depends on the following requirements:
- Nature and type of data to be collected
 - Business criticality of the data
 - Importance of the collected data
 - High availability and disaster recovery considerations for database processing
 - A database that addresses single points of failure
 - Intensity of the data communication
 - Integration with various sources of data for analytics

Characteristics of open source databases

An open source database for IoT should be fault-tolerant and highly

available. It should have the following characteristics:

- No vendor lock-in and should ensure seamless integration of enterprise-wide tools, applications, products and systems developed or deployed by different organisations and vendors.
- These databases should increase productivity, speed up time-to-market, reduce risks and increase quality.
- No vendor monopoly allows use of free and open source databases. With data transferability and open data formats, there are greater opportunities to share data across interoperable platforms.
- Adoption of open source databases enhances the interoperability with other enterprise applications because of reuse of software stacks, libraries, and components.

Layers of IoT architecture and the database they need

The architecture of an IoT database analytics system has the following significant requirements.

- **Context:** Capturing the context 24 hours a day, 365 days a year
- **Standards:** Leveraging standard protocols of communication between IoT devices and enterprise systems

- **Scalability:** Responding to increased load by declining performance, not failure; increasing capacity proportionally as resources are added
- **Data management:** Efficiently managing enormous volumes of data
- **Connectivity:** Providing high network connectivity for large data payloads and continuous streaming
- **Security:** Moving and encrypting information securely even as IoT introduces new risks and vulnerabilities

- **Interoperability:** Networking all systems together and ensuring interoperability of all data
- **Device management:** The ability to connect to devices remotely at scale to manage devices such as updating security credentials and firmware.
- **AI-driven analytics:** Enriching and exploring prescriptive and predictive analytics data to deliver actionable insights

In an enterprise IoT solution, thousands of sensors and actuators are connected to the edge server, continuously collecting data. The database must efficiently perform data transformation operations. The IoT data stream normalises the data to a standard format and sends it to a central repository.

The various layers of IoT architecture are data layer, things layer, communication layer, query layer, management layer, and application layer.

Things layer: This layer consists of entities that produce data for the IoT application and its modules. It covers IoT sensors, actuators, and devices that act like data production objects. Data collection, processing, and real-time data aggregation are performed at this layer. 'Data ingest' collects and stores logs and messages from the devices.